

A duplication-provenance $\times$ genomic-arrangement framework for paralogous transcription factors in the human immune system

iteration15 (human-only) — ETS P01 genomics platform

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Abstract

We classify every human paralogous transcription factor (TF) on a 2×2 of **duplication provenance** (2R whole-genome-duplication ohnolog vs. small-scale duplication, SSD) $\times$ **genomic arrangement** (clustered *cis* vs. dispersed), after removing the silent KRAB-ZNF and HOX arrays that confounded earlier analyses. Asking which of the four categories is enriched for cell-type *identity* markers versus cytokine-*response* genes, we find a clean dissociation: enrichment for lineage-identity markers is a specific property of *dispersed* ohnologs and is *erased* by clustering (PBMC odds ratio, OR= 1.79; spleen OR= 2.87; within-ohnolog clustered-vs-dispersed OR $\approx$ 0.33–0.47), whereas cytokine-response enrichment tracks ohnolog *provenance* and *survives* clustering. Dosage analysis shows buffering is not a category-level trait, but specific clustered neighbours diverge: SSD tandems co-amplify (SAND/SP100 family, buffering index B up to 1.84) while a *cis*-ohnolog pair, STAT1/STAT4, is the cleanest buffer ($B = 0.78$, partial $r = -0.41$). We interpret clustering as an evolutionary mechanism that *re-routes* the regulatory potential of a 2R duplicate—from divergent lineage specialisation toward a coordinated, dosage-coupled signalling-effector module, the configuration exemplified by the ETS (ETS1/FLI1/ETS2/ERG) and STAT paralogs.

1 Background and analytical framework

Gene duplication is the raw material of regulatory innovation, but not all duplicates are equivalent. Two axes matter. **Provenance**: a duplicate born in the two rounds of vertebrate whole-genome duplication (2R-WGD, ~ 500 Mya)—an *ohnolog*—is dosage-balanced and preferentially retained, whereas a small-scale duplicate (SSD; tandem or dispersed) is not. **Arrangement**: duplicates that remain in a genomic *cluster* share enhancers and topological context, while *dispersed* copies sit in independent regulatory landscapes. iteration15 crosses these into a single four-category label, `dup_class4`:

$$\{2R \text{ ohnolog}, SSD\} \times \{\text{clustered } cis, \text{dispersed}\} \rightarrow \{\text{dispersed_ohnolog}, \text{clustered_cis_ohnolog}, \text{clustered_SSD_tandem}, \text{dispersed_SSD}\}.$$

A decisive control precedes the analysis: the KRAB-ZNF and HOX arrays—large, silent, tandem-duplicated blocks shown previously to manufacture most of the raw “clustered” enrichment signal—are removed *up front*, so every result below is the KRAB/HOX-controlled effect.

2 Methods (summary)

Human TFs were curated from Lambert et al. (*Cell* 2018); families were defined by shared DNA-binding domain (DBD), with C2H2 zinc-fingers subdivided by HGNC accessory domain (KRAB/SCAN/BTB) and CSD RNA-binders removed. Members were placed on hg38 (GENCODE v45); KRAB-ZNF and HOX were excluded, and survivors were split into genomic clusters (two same-family members adjacent iff no *other* protein-coding gene lies between them) versus singletons. Ohnolog provenance

came from OHNOLOGS v2 plus a general *cis*-ohnolog rescue rule (any cluster containing ≥ 1 known ohnolog confers ohnolog status on all members), which recovers the textbook FLI1/ERG pair with no hand curation. Enrichment was tested per category by one-vs-rest Fisher exact tests (OR, 95% CI, BH-FDR) against the detectable paralogous-TF universe ($\text{catalog} \cap \text{scRNA-seq var_names}$), with pairwise contrasts and a provenance $\times$ arrangement factorial logit. Dosage buffering used the Oesinghaus et al. (2025) human cytokine dictionary: per within-family pair, buffering index $B = \text{Var}(x+y)/(\text{Var } x + \text{Var } y)$ and a common-mode-removed partial correlation, benchmarked against a cross-family null.

3 Part I–II: the four-category catalog

After KRAB/HOX removal, 1,163 paralogous TFs survive, forming 78 genomic clusters (184 clustered members) plus 979 singletons. Following pseudogene/readthrough cleanup, the classified universe is **1,153 TFs** (Table 1). The ETS quartet (ETS1/FLI1/ETS2/ERG) and the STAT family (STAT1/3/4/5A/5B) fall in **clustered_cis_ohnolog**; the chromosome-2 SAND cluster (SP100/110/140/140L), which contains no ohnolog member, remains **clustered_SSD_tandem**.

Table 1: The four duplication categories ($n = 1,153$ classified paralogous TFs).

Category	n	Reference members
dispersed_ohnolog	524	scattered 2R-WGD singletons
dispersed_SSD	450	scattered small-scale duplicates
clustered_cis_ohnolog	106	ETS1/FLI1/ETS2/ERG , STAT1/3/4/5A/5B
clustered_SSD_tandem	73	chr2 SAND: SP100/110/140/140L

4 Part III: identity vs. cytokine-response enrichment

4.1 Cell-type identity markers — a property of *dispersed* ohnologs

In both human PBMC and spleen, **dispersed_ohnolog** is the *only* FDR-significant identity-enriched group, and the two independent marker sets agree (Table 2, Figs. 1–2). Clustering abolishes the effect: within ohnologs, the clustered-vs-dispersed OR is 0.47 (PBMC, $p = 0.008$) and 0.33 (spleen, $p = 0.008$); both clustered categories collapse to the SSD background rate.

Table 2: Identity-marker enrichment (one-vs-rest Fisher). FDR by Benjamini–Hochberg.

Tissue	Category	rate	OR	FDR	
PBMC	dispersed_ohnolog	0.287	1.79	0.0002	enriched $\uparrow$
PBMC	dispersed_SSD	0.192	0.70	0.029	depleted
PBMC	clustered_cis_ohnolog	0.160	0.61	0.119	ns
PBMC	clustered_SSD_tandem	0.164	0.64	0.196	ns
Spleen	dispersed_ohnolog	0.163	2.87	7×10^{-7}	enriched $\uparrow$
Spleen	dispersed_SSD	0.063	0.42	0.0001	depleted
Spleen	clustered_cis_ohnolog	0.061	0.51	0.169	ns
Spleen	clustered_SSD_tandem	0.070	0.60	0.331	ns

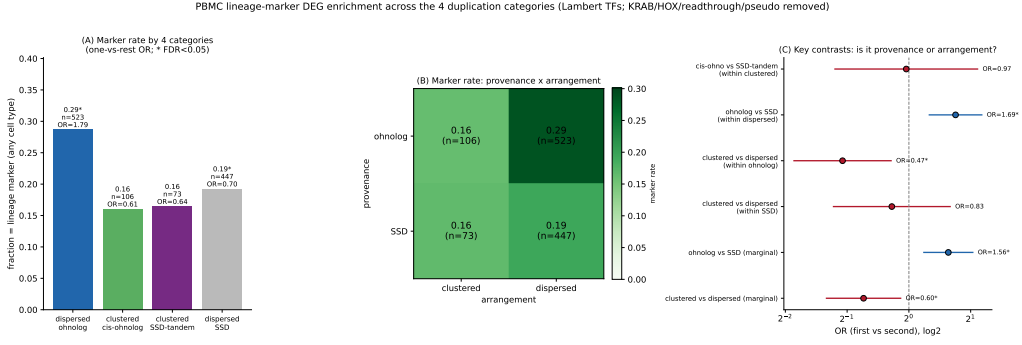


Figure 1: PBMC identity-marker enrichment across the four duplication categories. [dispersed_ohnolog](#) is the single enriched group.

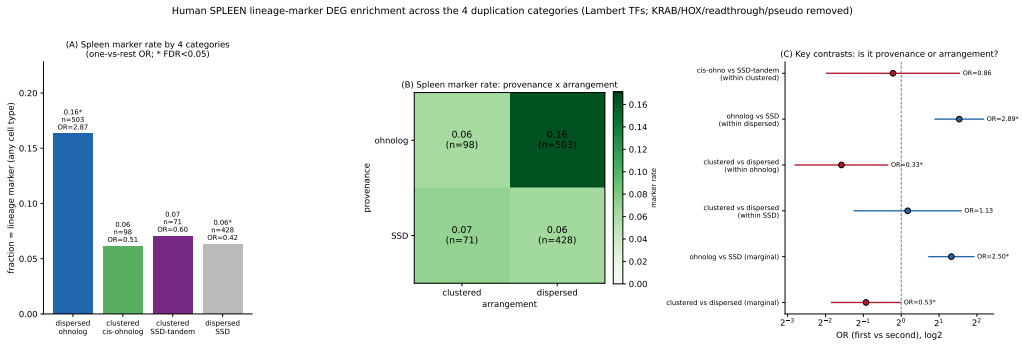


Figure 2: Spleen identity-marker enrichment (independent one-vs-rest markers) reproduces the PBMC pattern: enrichment is exclusive to [dispersed_ohnolog](#).

4.2 Cytokine-response genes — a property of ohnolog *provenance* that survives clustering

The cytokine-response picture dissociates sharply from identity (Table 3, Fig. 3). [dispersed_ohnolog](#) is again most enriched (rate 37.5%, OR= 1.67, FDR= 2.3×10^{-4}), but [clustered_cis_ohnolog](#) *stays high* (rate 29.2%, OR= 0.89, $p = 0.66$, i.e. not depleted), and the within-ohnolog clustered-vs-dispersed contrast is *not* significant ($p = 0.12$)—unlike identity. The provenance $\times$ arrangement factorial confirms the asymmetry: the ohnolog main effect is significant for all three readouts (PBMC adj-OR= 1.69; spleen 2.89; cytokine 1.67), the arrangement main effect and interaction are not.

Table 3: Cytokine-response enrichment (one-vs-rest Fisher).

Category	rate	OR	FDR	
dispersed_ohnolog	0.375	1.67	0.0002	enriched $\uparrow$
clustered_cis_ohnolog	0.292	0.89	0.661	retained (ns)
dispersed_SSD	0.264	0.68	0.007	depleted
clustered_SSD_tandem	0.219	0.59	0.119	ns

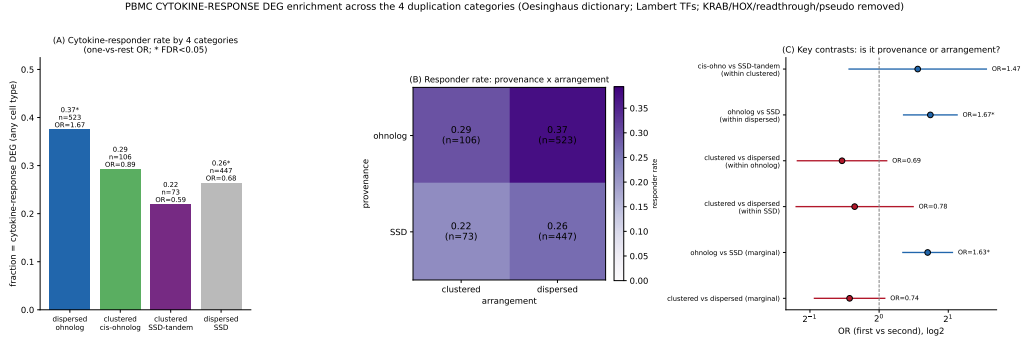


Figure 3: Cytokine-response enrichment. Unlike identity, the clustered *cis*-ohnolog group does not collapse—clustering preserves the cytokine programme.

5 Part IV: dosage buffering by category

Buffering is *not* a category-level trait—every category’s median $B \approx 1.0$, indistinguishable from the cross-family null (Kruskal–Wallis $p = 0.14$; Fig. 4). The signal lives in specific *neighbour* pairs, where the two clustered categories diverge (cis vs. SSD-tandem neighbour B contrast $p = 0.038$):

- **SSD-tandem neighbours co-amplify.** The chr2 SAND/SP100 family shows B up to 1.84 (SP100/SP110, partial $r = 0.82$); all six SAND pairs have $B = 1.4$ –1.8 with strong positive partial correlation—redundant gene-dose stacking.
- **A *cis*-ohnolog pair is the cleanest buffer.** STAT1/STAT4 has $B = 0.78$, and common-mode removal flips its correlation negative ($r = -0.27 \rightarrow$ partial $r = -0.41$): active compensation, not shared drive.

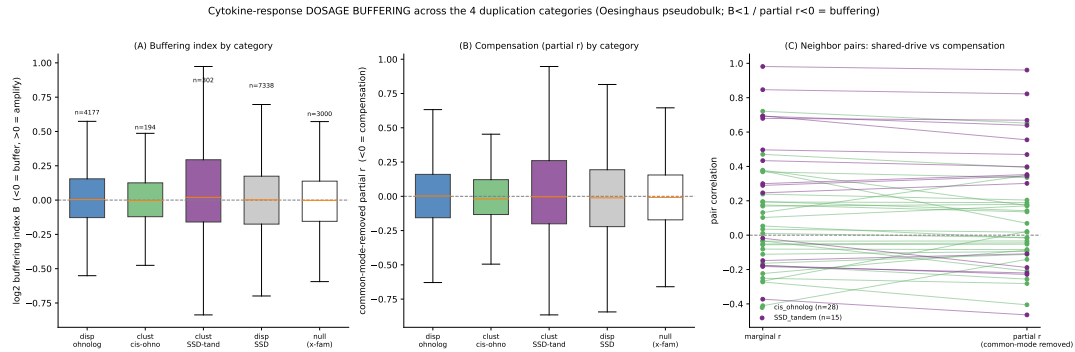


Figure 4: Cytokine-dictionary dosage buffering by category. Category medians sit on the null; divergence is a neighbour-level, pair-specific phenomenon (SAND amplification vs. STAT buffering).

6 Synthesis: what evolutionary advantage might clustering confer?

The central result is a **dissociation**: *provenance* (being a 2R ohnolog) sets a duplicate’s regulatory potential, but *arrangement* (clustered vs. dispersed) decides how that potential is spent. The two classical post-duplication fates—*neofunctionalisation* and *dosage/sub-functional sharing*—map cleanly onto the two arrangements, and clustering appears to carry several distinct selective advantages.

1. Dispersal buys divergence; clustering buys coordination. A dispersed ohnolog sits in its own regulatory landscape and is free to acquire lineage-specific enhancers—hence [dispersed_ohnolog](#) is the identity-TF reservoir. Clustering removes that freedom: shared enhancers and a shared topological domain constrain the copies to fire together. The advantage of *losing* divergence is **guaranteed co-regulation**: a cluster behaves as one regulatory unit, ensuring its members are co-expressed in the right cell at the right time. This is exactly what we observe—clustered ohnologs lose the (divergence-dependent) identity signal but *retain* the (coordination-dependent) cytokine-response signal.

2. Clustering converts an identity candidate into a signalling effector. The same [clustered_cis_ohnolog](#) members that are *not* enriched for identity markers *are* enriched (or at least not depleted) for cytokine-response genes. We read this as a functional re-assignment: a 2R ohnolog that might otherwise have become a divergent lineage-determining factor is instead recruited, as a coordinated block, into a fast, signal-responsive effector programme. The ETS (ETS1/FLI1/ETS2/ERG) and STAT paralogs—the engines of antigen-receptor and cytokine signalling—are the archetypes. The selective advantage is a **ready-made, dosage-matched module** that can be deployed coherently downstream of a signal.

3. Clustering enables paralog buffering at dosage-sensitive nodes. True compensatory buffering (STAT1/STAT4: partial $r = -0.41$) is found in a *cis*-ohnolog pair, consistent with the idea that physical proximity plus shared ancestry maintains functional redundancy precisely where dosage matters. Buffering is selectively valuable because it makes the network **robust to perturbation of a single member**: if one paralog is lost or mutated, the neighbour compensates. This is the molecular basis for the ETS P01 thesis—a paralog-buffered network whose phenotype on perturbation depends on how much redundancy remains.

4. Clustering can also stack dose, not just buffer it. The SSD-tandem SAND/SP100 family shows the opposite neighbour behaviour—co-amplification ($B > 1$). Here the advantage of physical clustering is **additive dosage**: tandem copies of the same response amplify output, the simplest adaptive payoff of a local duplication, with no regulatory rewiring. Thus clustering is not a single strategy; its consequence (buffer vs. amplify) is itself gated by provenance (ohnolog vs. SSD).

In one sentence: a 2R-WGD origin gives a TF pair the *capacity* to become either a divergent identity factor or a coordinated effector, and genomic clustering is the switch—trading the evolvability of dispersal for the robustness and coordination of a co-regulated, dosage-coupled module.

7 Caveats

1. The OR denominator is the detectable paralogous-TF universe, not an expression-matched background; the FDR-backed claims are the [dispersed_ohnolog](#) identity enrichment and the provenance factorial. The within-clustered contrasts rest on small groups ($n \approx 73$ –106) and are trends.
2. Spleen cell-type labels derive from a blood-PBMC Azimuth transfer, so spleen is a reproducibility contrast rather than added cell-type diversity; the cytokine dictionary is also PBMC.
3. Provenance depends on the conservative OHNOLOGS v2; the *cis*-ohnolog rescue fixes only clustered misses (dispersed misses such as FEV stay SSD). The four-category `dup_class4` label—not any per-gene mode—is the unit of analysis.

8 Conclusion

Across human PBMC and spleen identity programmes and a PBMC cytokine dictionary, 2R-WGD ohnolog *provenance* is the determinant of a paralogous TF’s regulatory enrichment, while genomic *arrangement* re-routes it: dispersal toward divergent cell-identity roles, clustering toward coordinated, dosage-coupled cytokine effectors with the capacity for paralog buffering. Clustering’s evolutionary advantage is therefore not a single thing but a trade—coordination, modular deployability and robustness in exchange for the evolvability of independent divergence—and the ETS and STAT paralogs are its clearest immune-system instantiations.